STA2002 Probability and Statistics II
Assignment 3
Due date: **5 November 2023 11:59pm**

Please work on the assignment by yourself only. We follow a strict rule on academic honesty.

1. According to a population census in 1986, the percentage of males who are 18 or 19 years old and are married was 3.7%. We shall test whether this percentage increased from 1986 to 1988.

   (a) Define the null and alternative hypotheses.

   $$H_0 : p = 0.037, H_1 : p > 0.037.$$

   (b) Define a critical region that has an approximate significance level of $\alpha = 0.01$.

   $$t = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \geq 2.326.$$

   (c) If $y = 20$ out of a random sample of $n = 300$ males, each 18 or 19 years old, were married (*U.S. Bureau of the Census, Statistical Abstract of the United States: 1988*), what is your conclusion?

   $$t = 2.722 > 2.326, \text{reject } H_0.$$

2. Michael suspects that the quarter in his pocket may not be a fair coin. He flips it 50 times, and to conduct a two-sided hypothesis test at the $\alpha = 0.05$ significance level. Heads appears $h = 30$ times.

   (a) Does Michael reject H_0?

   Let p be the probability that the coin comes up heads. Michael is conducting the hypothesis test

   $$H_0 : p = 0.5 \qquad\qquad H_1 : p \neq 0.5.$$

   Here, $n = 50$ and $\bar{x}_{50} = 30/50 = 0.6$, from which we obtain the test statistic

   $$t = \frac{0.6 - 0.5}{\sqrt{(0.5)(0.5)/50}} \approx 1.414$$

   and the p-value

   $$p = P(Z \leq -1.414) + P(Z \geq 1.414) \approx 0.157.$$

   Michael fails to reject H_0. Alternatively, since

   $$|t| = 1.414 < z_{0.025} = 1.96,$$

   Michael fails to reject H_0.

(b) For what values of h would Michael reject H_0, while fixing $n = 50$?

Michael would reject H_0 if either $t \geq z_{0.025} = 1.96$ or $t \leq -z_{0.025} = -1.96$. Both inequalities hold when

$$|t| = \left| \frac{\overline{x} - 0.5}{\sqrt{0.5(0.5)/50}} \right| \geq 1.96$$

Or, equivalently

$$(\overline{x} - 0.5)^2 \geq 1.96^2 \frac{0.25}{50}$$

Solving the quadratic equation

$$(x - 0.5)^2 - \frac{1.96^2}{250} = 0$$

gives the solutions $x = 0.376$ and $x = 0.624$. Thus, we must have that $t \geq 1.96$ when $\overline{x} \geq 0.624$, and $t \leq -1.96$ when $\overline{x} \leq 0.376$. Since sample means of 0.351 and 0.648 correspond to obtaining roughly 18.8 and 32.75 heads, respectively, Michael would reject H_0 if either $h \leq 18$ or $h \geq 33$.

3. (Sample size determination for hypothesis testing) Suppose that $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$ and σ^2 is known. We would like to test

$$H_0 : \mu = \mu_0, \quad H_1 : \mu > \mu_0.$$

We consider the test statistic

$$T = \overline{X}$$

and the critical region at significance level α

$$C = \left\{ \overline{x};\ \overline{x} > \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}} \right\}.$$

(a) We write $\beta(\mu')$ to be the probability of type II error given the true value of $\mu = \mu'$. Prove that

$$\beta(\mu') = \Phi\left(z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}} \right),$$

where $\Phi(z) = P(Z \leq z)$ is the cdf of $Z \sim N(0, 1)$.

Assuming the true value of $\mu = \mu'$, $\overline{X} \sim N(\mu', \sigma^2/n)$, and so

$$\begin{aligned} \beta(\mu') &= P\left(\overline{X} \leq \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}} \right) \\ &= P\left(\frac{\overline{X} - \mu'}{\sigma/\sqrt{n}} \leq z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}} \right) \\ &= \Phi\left(z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}} \right). \end{aligned}$$

(b) We would like to determine the sample size n so that the probability of type I error is α and the probability of type II error is β. Using part (a), prove that

$$n = \left(\frac{\sigma (z_\alpha + z_\beta)}{\mu_0 - \mu'} \right)^2.$$

We set

$$\beta = \Phi\left(z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}} \right),$$

and this leads to

$$-z_\beta = z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}.$$

Rearranging gives

$$n = \left(\frac{\sigma (z_\alpha + z_\beta)}{\mu_0 - \mu'} \right)^2.$$

(c) Let μ denote the true average tread life of a certain type of tire. Consider testing $H_0 : \mu = 24,000$ versus $H_1 : \mu > 24,000$ from a normal population distribution with $\sigma = 1300$. Assuming the true value of μ is $\mu' = 25,000$. What is the sample size n so that the probability of type I error is $\alpha = 0.025$ and the probability of type II error is $\beta = 0.05$?

First, note that

$$z_{0.025} = 1.96, \quad z_{0.05} = 1.65,$$

and so

$$n = \left(\frac{1300(1.96 + 1.65)}{24,000 - 25,000} \right)^2 = (-4.693)^2 = 22.02.$$

$n = 23$ is needed to achieve the desired type I and type II error.

4. (A/B testing) When designing websites, people would like to design a website that maximize the click through rate (CTR), that is, the probability that an user will click on an advertisement presented on the website. This is usually done by what people call A/B testing. Suppose that we have two designs of the website, say Version A and Version B. Denote by p_A (respectively p_B) to be the proportion of users that click on an advertisement in Version A (respectively Version B) of the website. Use your favourite computing language to answer this question, and attach your code at the end of your assignment.



(a) We would like to test whether the proportion of users that click on an advertisement is different between Version A and Version B. Formulate the null hypothesis H_0 and the (two-sided) alternative hypothesis H_1.

$$H_0 : p_A = p_B, \quad H_1 : p_A \neq p_B.$$

(b) Download the dataset "AB_2020.csv" on Blackboard. There are 4 columns with 294490 data points. We will only use the column "landing_page" and "converted". "landing_page" is either "A" or "B", indicating the user is landing on Version A or Version B. "converted" is either "1" or "0", with "1" meaning the user clicks on an advertisement and "0" meaning the user does not click on an advertisement.

Report $\widehat{p}_A$ and $\widehat{p}_B$, the maximum likelihood estimate of p_A and p_B respectively.

$$\begin{aligned}
n_A &= 147245 \\
n_B &= 147245 \\
y_A &= 17740 \\
y_B &= 17500 \\
\widehat{p}_A &= \frac{17740}{147245} = 0.1205 \\
\widehat{p}_B &= \frac{17500}{147245} = 0.1188 \\
\widehat{p} &= \frac{17740 + 17500}{294490} = 0.1197
\end{aligned}$$

(c) Assume that the user behaviour in Version A is independent of that in Version B. Compute the test statistic of the hypothesis test.

$$t = \frac{0.1205 - 0.1188}{\sqrt{0.1197(1 - 0.1197)(1/147245 + 1/147245)}} = 1.4210.$$

(d) Is H_0 rejected at 5% significance level? That is, are the two versions of the website really give different CTR at 5% significance?

Since

$$|t| = 1.4210 < 1.96 = z_{0.025},$$

we fail to reject H_0 at 5% significance.